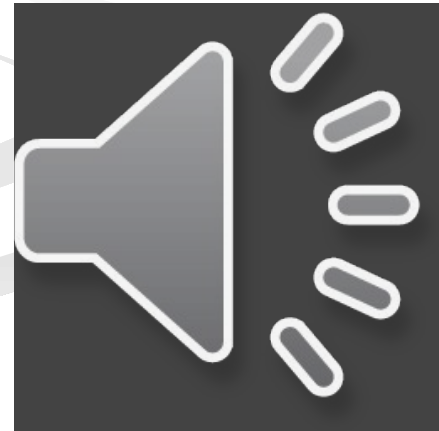


Meta-analysis

**Master of Statistics
KU Leuven**

Study retrieval



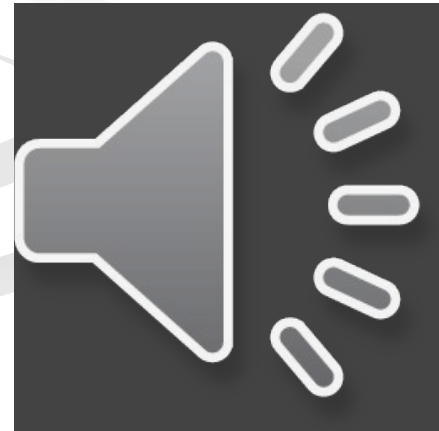
Steps in a meta-analysis

1. Formulating research questions
2. Systematic search for studies
3. Selection of relevant studies
4. Extracting relevant data (outcomes, study characteristics, study quality)
5. Converting data to a common effect size measure
6. Combining and comparing the effect sizes
7. Looking for moderator variables
8. Concluding and reporting



Outline

- a. Problem formulation**
- b. Searching for studies
- c. Selecting studies
- d. Issue of quality



Problem formulation

As in other studies, a clear problem formulation is needed (although it might still change somewhat, for instance when finding too much or too few studies). This includes information on:

- Population
- Variables
- Hypotheses/research questions
- Further demarcation of the problem

e.g.: “*relation between challenging behaviour of persons with mental retardation and behaviour of staff*” = may be too vague & broad

→ Better: “*relation between self-injury of persons with severe mental retardation and stimulating behaviour of staff*”



Helpful rule: PICO-model

A good research question often includes the following elements:

- P Population
- I Intervention/interest
- C (Comparison)
- O Outcome



Helpful rule: PICO-model

E.g.: “What is the effect on drug abuse in secondary school students of giving information at school rather than giving sanctions to drug users”

- P Population: **secondary school students**
- I Intervention/interest: **giving information at school**
- C (Comparison): **giving sanctions to drug users**
- O Outcome: **drug abuse**



Outline

- a. Problem formulation**
- b. Searching for studies**
- c. Selecting studies
- d. Issue of quality



Systematic search: Thorough and transparent search for all relevant (published and unpublished) study results

- a) **Articles**
- b) **Books**
- c) **Conference papers**
- d) **Dissertations**
- e) **(Un)finished research reports**
- f) **...**



Retrieving literature

1. Using (online) searches
2. Browsing secondary sources
3. Citation searches
4. Contacting researchers
5. Browsing journals
6. Research registers



Retrieving literature

1. Using (online) searches: ask librarian!

<https://bib.kuleuven.be/english/training-and-tutorials>

- a) **Databases (ERIC, PsycINFO, PubMed, Dissertation Abstracts International, Web of Science, ...**
- b) **Catalogue library (KU Leuven: Limo)**
- c) **Google, Google Scholar (<http://scholar.google.com/>), Altavista, Yahoo**
...



Some major databases

- **PubMed (Medline):** PubMed was developed by the National Center for Biotechnology Information (NCBI). It was developed in conjunction with publishers of biomedical literature as a search tool for accessing literature citations and linking to full-text journals at web sites of participating publishers.
- **PsycINFO:** References and abstracts of journal articles, books and book articles concerning psychology and its related disciplines (medicine, psychiatry, nursing, sociology, educational studies, pharmacology, physiology, anthropology, law, linguistics). The archive starts from 1887. PsycINFO contains also psychological dissertations from Dissertation Abstracts International.
- **ERIC**, the world's largest index to journal articles and research reports in education, includes citation and abstract information from over 750 educational journals and related documents, in total more than 1 million references. Also gives access to more than 100 000 full-text documents (research reports, books, conference papers AERA, ...)
- Other ... Combine databases!





Database Search

Eric



0-9 A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

OTHERS

5 Results

Tweak my results

Sort by Relevance ▾

Search Also ^



Databases by category

- > Archeology & Art Sciences
- > Architecture
- > Audiovisual & Visual Arts
- > Bioscience
- > Catalogues & Reference Works
- > Economics & Business Administration
- > Engineering
- > Health Sciences
- > History
- > Language & Literature

1 **Eustathius of Thessalonica's Commentary on the Odyssey**

Cullhed, Eric [editor] ; Olson, S. Douglas [editor] ; Eustathius (Archbishop of Thessalonica,) (-approximately 1194) [author]
This online publication offers a new text edition of the Byzantine scholar and rhetorician Eustathius of Thessalonica's Commentary on the Odyssey, composed during the latter half of the twelfth century CE. The Commentary collects material from a wide range of different sources which explain or expand on words, phrases and ideas in the Homeric epic. Original comments are blended with extracts from earlier commentators, especially the Homeric scholia. The text is an important source for fragments of otherwise lost works of ancient literature, for the history of exegesis and lexicography, and for Byzantine cultural history. This is the first complete critical edition of the text, and the first translation of it into a modern language.

[Online access >](#)2 **Educational Resources Information Center / ERIC (OvidSP)**

Bevat Amerikaanse pedagogische en onderwijskundige literatuur (tijdschriftartikels, rapporten, boeken en conferentievoordrachten (o.a. AERA papers). Meer dan 1 miljoen referenties. Geeft ook gratis toegang tot meer dan 100.000 full-text documents (research reports, speeches, meeting papers AERA,...).
ERIC, the world's largest index to journal articles and research reports in education, includes citation and abstract information from over 750 educational journals and related documents.

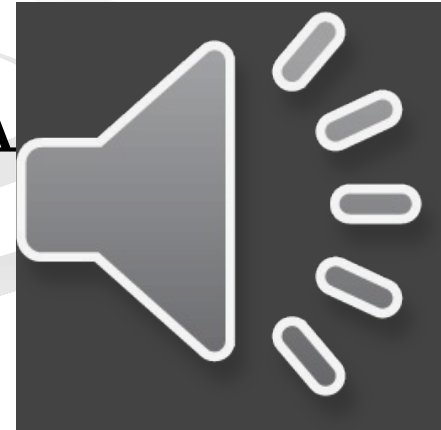
[Online access >](#)3 **ERIC - Education Resources Information Center (USDE)**

Bevat Amerikaanse pedagogische en onderwijskundige literatuur (tijdschriftartikels, rapporten, boeken en conferentievoordrachten (o.a. AERA papers). Meer dan 1 miljoen referenties. Geeft ook gratis toegang tot meer dan 100.000 full-text documents (research reports, speeches, meeting papers AERA,...).
The Education Resources Information Center (ERIC) is a digital library of education-related resources.

[Online access >](#)

Searching relevant literature in databases

- Analyse your question in *keywords*, *search descriptors*, or *search terms*
- Combine keywords to refine search:
 - “search a complete string”
 - Truncating: Meta-analy* (includes meta-analysis, meta-analytical, meta-analyses, ...)
 - Boolean operators: OR, AND, NOT, NEAR/4
- Search for these terms in titles keywords [AK] and abstracts [A topic: [TS])



An example:

*(disability OR mental retardation OR complex needs OR high support needs OR cognitive impairment OR learning difficulty) AND
(challenging behaviour OR problem behaviour OR aggression OR self-injury OR self-injurious behaviour) AND
(staff OR carer OR care worker OR special educator)*

Note: also singular & plural; US and UK notation!

Lambrechts, G., Van Den Noortgate, W., Eeman, L., Maes, B. (2010). Staff reactions to challenging behaviour: An observation study. *Research in Developmental Disabilities*, 31 (2), 525-535.



An example:

*(disability OR mental retardation OR complex needs OR high support needs OR cognitive impairment OR learning difficulty) AND
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Lambrechts, G., Van Den Noortgate, W., Eeman, L., Maes, B. (2010). Staff reactions to challenging behaviour: An observation study. *Research in Developmental Disabilities*, 31 (2), 525-535.



A second example:

(Misconception OR error OR misuse OR misinterpretation OR misunderstanding OR fallacy) AND (inference OR sampling OR normal OR confidence OR interval OR test OR level OR p-value) AND (course OR college OR university)

Castro Sotos, A.E., Vanhoof, S., Van Den Noortgate, W., Onghena, P. (2007). Students' misconceptions of statistical inference: A review of the empirical evidence from research on statistics education. *Educational Research Review*, 2, 98-113.



Basic Search | Find Citation | **Search Tools** | Search Fields | Advanced Search | Multi-Field Search1 Resource selected | [Hide](#) | [Change](#)

ERIC 1965 to January 2022

Tool Type Thesaurus ▼ misconceptions

Search

Use the thesaurus to
find other relevant
keywords

✓ Misconceptions		8505	☐	☐	
[Used For]					
	Mistaken Conceptions				
	Myths (Misconceptions)				
[Related Terms]					
☐	Cognitive Structures	2631	☐	☐	
☐	Comprehension	6342	☐	☐	
☐	Concept Formation	17392	☐	☐	
☐	Concept Mapping	2831	☐	☐	
☐	Learning Processes	26802	☐	☐	
☐	Mathematical Concepts	10959	☐	☐	
☐	Scientific Concepts	16141	☐	☐	



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| ☐ fl IES Funding Link | ☐ pl IES Publication Link | ☐ ib ISBN | ☐ is ISSN |
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| ☐ jx Journal Word | ☐ lg Language | ☐ li Law Identifiers | ☐ lv Level of Availability |
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| ☐ sa Sponsoring Agency | ☐ tw Text Word | ☒ ti Title | ☐ ur URL |
| ☐ vo Volume | ☐ wl WWC Study Review Guide Link | ☐ wc What Works Clearinghouse Reviewed | ☐ yr Year of Publication |



☐ 1. **The Effects of Attentional Focus on Test-Retest Reliability of Jumping Tasks**

[Abstract Reference](#)
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 **KU Leuven**

Makaruk, Hubert. Porter, Jared M. Cieslinski, Igor.

Measurement in Physical Education and Exercise Science. v25 n4 p306-313 2021

URL

<http://dx.doi.org/10.1080/109136...>

[► Abstract](#) [Cite](#) [+ My Projects](#) [+ Annotate](#)

☐ 2. **Mapping Pre-Service Teachers' Faulty Reasoning in Geometric Translations to the Design of Van Hiele Phase-Based Instruction**

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[ERIC Full-text](#)

Mbus, Nokwanda P. Luneta, Kakoma.

South African Journal of Childhood Education. v11 n1 Article 871 2021

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Kurniati, Utut. Setiyani. Sagita, Laela.

Anatolian Journal of Education. v6 n2 p135-156 Oct 2021

[► Abstract](#) [Cite](#) [+ My Projects](#) [+ Annotate](#)

Link to full text



Remark:

Trade off between *recall* (proportion of relevant literature that is found) and *precision* (proportion of retrieved literature that is retained for the meta-analysis).



Retrieving literature

1. Using (online) searches
2. Browsing secondary sources
 - a) **Encyclopedias**
 - b) **Special journals (Psychological Bulletin, Educational Research Review, ...)**
 - c) **Reviews in other journals**
 - d) **Handbooks**
 - e) **Conference programs & proceedings**
 - f) **...**



Retrieving literature

1. Using (online) searches
2. Browsing secondary sources
3. Citation searches based on relevant work
 - a) **Backward: browsing reference lists**
 - b) **Forward: Who cited this work?**



☐ 1. **The Effects of Attentional Focus on Test-Retest Reliability of Jumping Tasks**

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► Abstract  Cite  + My Projects  + Annotate

Abstract Reference
Complete Reference

 Find Similar
 Find Citing Articles

 KU Leuven

ERIC Full-text

Forward citation search



Retrieving literature

1. Using (online) searches
2. Browsing secondary sources
3. Citation searches
4. Contacting researchers
 - a) **Informal conversations at conferences**
 - b) **Informal conversations with colleagues or professors**
 - c) **Electronic discussion lists**
 - d) **Formal request to key researchers**
 - e) ...

Look for heterogenous contacts!



Retrieving literature

1. Using (online) searches
2. Browsing secondary sources
3. Citation searches
4. Contacting researchers
5. Browsing journals
 - a) **Use online tables of contents**
 - b) ...



Retrieving literature

1. Using (online) searches
2. Browsing secondary sources
3. Citation searches
4. Contacting researchers
5. Browsing journals
6. Research registries

Esp. Governmental organizations: FDA, IES,

For a list of registers of clinical research and clinical trials, see the appendix of

Dickersin, K. (1994). Research registers. In H. Cooper, & L. V. Hedges (Eds.), *The handbook of research synthesis* (pp. 71-83). New York: Sage



Note:

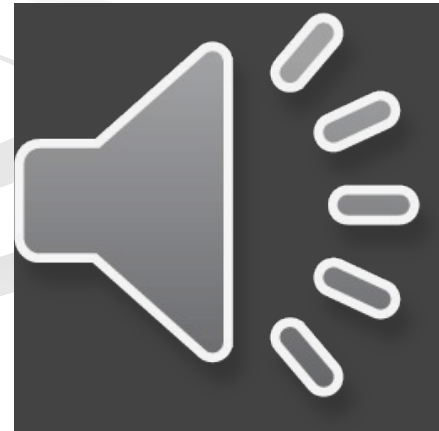
Some subpopulations, interventions, outcomes or designs that are relevant might not have been studied (sufficiently)!

- ⇒ Pay attention when interpreting/concluding
- ⇒ Suggest directions for further research!



Outline

- a. Problem formulation**
- b. Searching for studies**
- c. Selecting studies**
- d. Issue of quality



Inclusion & exclusion criteria

- Characteristics of population (e.g., age)
- Treatment type or duration
- Design features (e.g., presence of control group or control condition)
- Study quality (e.g., double blind RCT; trained observers, ...)
- Sufficient information about the outcome or about other study characteristics
- ...



Inclusion & exclusion criteria

- Make and report an explicit list (consistency and replicability are important quality criteria)!
- Narrow criteria will lead to a relatively small and homogeneous set of studies
 - Easy interpretation
 - ☞ Power and accuracy
 - ☞ Generalizability?
- Possible approach: start with somewhat broader criteria, that are refined after a pilot search



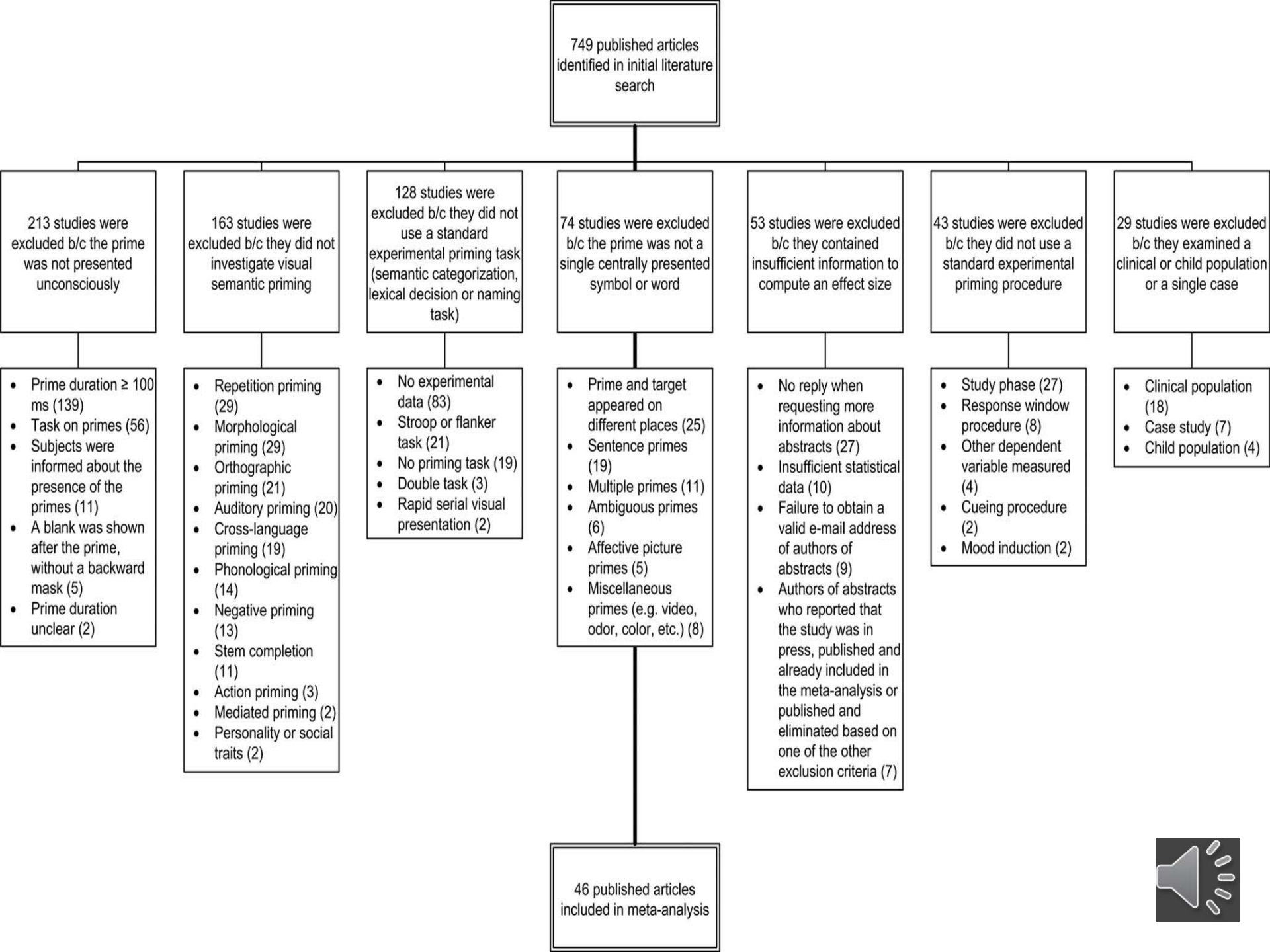
An example

**Van den Bussche, E., Van den Noortgate, W., & Reynvoet, B. (2009).
Mechanisms of masked priming: A meta-analysis. *Psychological Bulletin*, 135, 452-477.**

We used the following criteria to select studies for inclusion in the meta-analysis:

- 1. The relation between the prime and target should be of a direct semantic nature (for example, cat-dog) in the visual domain. Studies investigating phonological priming, morphological priming, [...] were excluded*
- 2. The primes have to be presented subliminally (i.e. beneath the consciousness threshold). [...]*
- 3. This meta-analysis focuses solely on studies employing the most standard experimental tasks in priming: [...]*
- 4. We limited the present meta-analysis to centrally presented single word or symbol primes. [...]*
- 5. We focused exclusively on studies that tested a healthy sample. [...]*
- 6. Finally, only studies were included which contained sufficient statistical information to compute an effect size. [...]*





Outline

- a. Problem formulation**
- b. Searching for studies**
- c. Selecting studies**
- d. Issue of quality**



The quality of the MA depends on the quality of the primary studies

But...

- Quality is a continuum, not a dichotomy
- Problem of judging quality: no consensus on what quality is
- Several criteria can (and should) be used
- Calculation of a (weighted) average of the criteria? Weights?
- Problem that data to judge quality not always reported (e.g., whether assumptions underlying the statistical analysis were checked; whether missingness was accounted for; ...)
- Problem of efficiency (if you end up with a few studies, power and accuracy is small)
- Problem of bias (e.g., some questions are difficult to study with an experimental designs)



Illustration of lack of consensus

Evidence-based Practice Special Interest Group. (2007). What counts as evidence?
Newsletter of the Evidence-Based Practice Special Interest Group, 1, 6. Retrieved on
 2008, February 14 on
http://www.abainternational.org/Special_Interests/EBPSIG0701c.pdf.

Below is a summary of how several groups concerned with evidence-based practice have rated single-subject experimental designs (SSDs) as a form of evidence.

SSDs are...	Discipline		
	Education	Psychology and Related Human Services	Medicine
Persuasive evidence	National Institute for Literacy ¹ What Works Clearinghouse ²	Society for Clinical Psychology ⁶	
Modest evidence	Campbell Collaborative ³ White House Office of Mgmt and Budget ⁴		Joanna Briggs Institute ⁹ Centre for Evidence-Based Medicine ¹⁰
Uninformative (or not mentioned)	U.S. Department of Education ⁵	Rand Promising Practices Network ⁷ Centre for Evidence-Based Mental Health ⁸	Cochrane Collaboration ¹¹ American Academy of Neurology ¹² Agency for Healthcare Research Quality ¹³



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Critical Appraisal Checklists

We offer a number of **free downloadable checklists** to help you more easily and accurately perform critical appraisal across a number of different study types.

The CASP checklists are easy to understand but in case you need any further guidance on how they are structured, take a look at our [guide on how to use our CASP checklists](#).

- > [Systematic Reviews with Meta-Analysis of Observational Studies](#)
- > [Systematic Reviews with Meta-Analysis of Randomised Controlled Trials \(RCTs\)](#)
- > [Randomised Controlled Trial \(RCT\) Checklist](#)
- > [Systematic Review Checklist](#)
- > [Qualitative Studies Checklist](#)
- > [Cohort Study Checklist](#)
- > [Diagnostic Study Checklist](#)
- > [Case Control Study Checklist](#)

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<https://sites.google.com/site/riskofbiastool/welcome>



Risk of bias
tools

^ Welcome

▼ RoB 2 tool

▼ ROBINS-I tool

ROBINS-I V2 tool

ROBINS-E tool

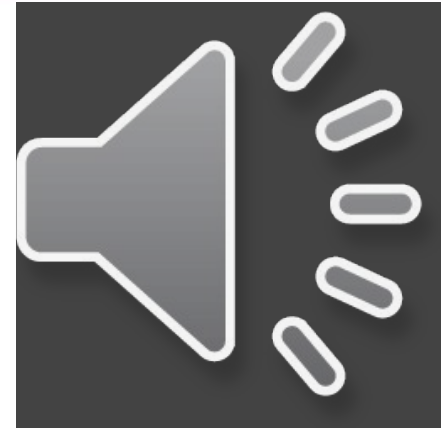
ROB-ME tool

robvis (visualization tool)

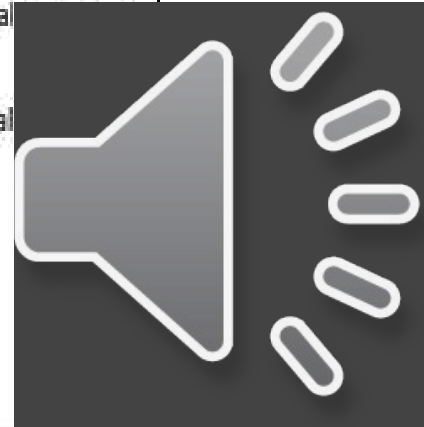
riskofbias.info

Welcome to our pages for risk of bias tools for use in systematic reviews.

- [RoB 2 tool \(revised tool for Risk of Bias in randomized trials\)](#)
- [ROBINS-E tool \(Risk Of Bias in non-randomized Studies - of Exposures\)](#)
- [ROB ME \(Risk Of Bias due to Missing Evidence in a synthesis\)](#)
- [ROBINS-I tool \(Risk Of Bias in Non-randomized Studies - of Interventions\)](#)
- [robvis \(visualization tool for risk of bias assessments in a systematic review\)](#)



Signalling questions	Elaboration	Response options
1b.1 Were all the individual participants identified and recruited (if appropriate) before randomization of clusters?	Answer 'Yes' if: (1) all participants were identified and recruited before the clusters were randomized; or (2) individual participants were not recruited at all but all were identified before randomization. In these cases identification/recruitment bias is not possible. Answer 'No' if: (1) some or all participants were identified or recruited after randomization; or (2) there are any clusters in which no participants were recruited (empty clusters).	<u>Y</u>/PY/PN/<u>N</u>/NI
1b.2 If N/PN/NI to 1b.1: Is it likely that selection of individual participants was affected by knowledge of the intervention assigned to the cluster?	Answer 'Yes' if: (1) those recruiting individuals were aware of cluster allocation before recruitment and this is likely, consciously or subconsciously, to have affected recruitment differentially between the intervention groups; (2) some participants were aware of cluster allocation before their recruitment and this is likely to have affected recruitment differentially between the intervention groups; or (3) those identifying potential participants (when recruitment is to take place subsequently) are aware of cluster allocation and are likely, consciously or subconsciously, to have differentially included potential individual participants in different trial groups or those identifying actual participants (when there is no subsequent recruitment) are aware of cluster allocation and are likely consciously or subconsciously, to have differentially included potential participants in different trial groups. Answer 'No' if all of the following (as relevant depending on the trial) are unaware of cluster allocation at recruitment: (1) those identifying actual participants, (2) those identifying potential participants; (3) those recruiting; and (4) potential participants.	NA/<u>Y</u>/PY/PN/<u>N</u>/NI



Some study quality criteria

- Construct validity
 - No 'blinding' of patients or of experimenters, inadequate operationalisation of treatment or placebo, ...
- External validity
 - E.g., overgeneralisation of results to other groups of participants, time periods or contexts
- Internal validity
 - E.g., history, maturation, testing, selection effects, mortality,... not under control
- Statistical conclusion validity
 - E.g., violated assumptions or wrong statistical test, capitalisation on chance, heterogenous sample, ...

How to deal with quality?

- Including only best studies?
- Excluding only very worst studies?
- Weighting according to quality?
- Investigating effect of quality?
 - by including quality as a predictor variable
 - by performing a sensitivity analysis

Study quality: guidelines

- Do not use low quality studies
- Use several indices for quality
- Look for the effect of designs
- Train coders and assess the reliability of coding!
- Report your quality criteria
- Peer reviewing is a first guarantee for quality, but ...

Inter- & intracoder reliability of search and selection process

- Unambiguous in- and exclusion criteria (internal reliability or consistency). E.g., how is good quality defined?
- Report important choices (external reliability or replicability)
- Coders should be trained
- Selection preferably (partly) done by two independent coders
- Report reliability of selection process

